



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE
Batch : 2025-29

Assignment : 17

Subject: Programming with Java

ASSIGNMENT No : 17

TITLE :Write a Java program to demonstrate the use of Swing and Applets

Problem Statement

Write a Java program to demonstrate the use of Swing and Applets

Learning Objectives

To develop graphical user interfaces using Swing components and Applets.

Theory

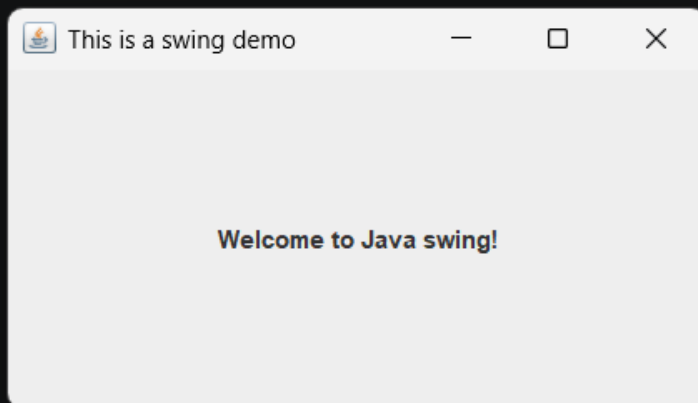
Swing is Java's GUI toolkit that provides lightweight components such as buttons, labels, text fields, and menus. Applets were small Java programs embedded in web pages for interactive applications. Although Applets are obsolete, Swing remains an important framework for desktop application development. Swing supports platform-independent graphical interfaces.



Code and Output Screenshots

1) Java Swing

```
J Javaswng.java > Java >  Javaswng
1  import javax.swing.*;
2
3  public class Javaswng {
    Run main | Debug main | Run | Debug
4      public static void main (String[] args) {
5          JFrame frame = new JFrame("This is a swing demo");
6          JLabel label = new JLabel("Welcome to Java swing!", JLabel.CENTER);
7
8          frame.add(label);
9
10         frame.setSize(350, 200);
11         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
12         frame.setVisible(true);
13     }
14 }
```





2) Java Applet : (The type Applet has been deprecated since version 9 and marked for removal)

```
J Javaplt.java > Java > Javaplt
1  import java.applet.*;
2  import java.awt.*;
3
4  public class Javaplt extends Applet {
5      public void paint (Graphics g) {
6          g.drawString("Welcome to Java Applet", 50, 50);
7      }
8  }
```

Conclusion

Thus, the program successfully demonstrates the development of graphical user interfaces (GUIs) using Java Swing components and Applets. It helps in understanding how graphical elements can be created and displayed in Java applications.



Exercise

Q1) Create a simple Student Registration Form GUI using Swing.

Sol) Code :

```
J Javaswng.java > Java >  Javaswng
1  import javax.swing.*;
2  public class Javaswng {
    Run main | Debug main | Run | Debug
3      public static void main(String[] args) {
4
5          JFrame f = new JFrame("Student Registration Form");
6
7          JLabel name = new JLabel("Name:");
8          JLabel roll = new JLabel("Roll No:");
9          JLabel course = new JLabel("Course:");
10
11         JTextField n = new JTextField();
12         JTextField r = new JTextField();
13         JTextField c = new JTextField();
14
15         JButton btn = new JButton("Register");
16
17         name.setBounds(40, 40, 80, 30);
18         n.setBounds(120, 40, 200, 30);
19
20         roll.setBounds(40, 90, 80, 30);
21         r.setBounds(120, 90, 200, 30);
22
23         course.setBounds(40, 140, 80, 30);
24         c.setBounds(120, 140, 200, 30);
25
26         btn.setBounds(120, 190, 100, 30);
27
28         f.add(name);
29         f.add(n);
30         f.add(roll);
31         f.add(r);
32         f.add(course);
33         f.add(c);
34         f.add(btn);
35
36         f.setSize(380, 280);
37         f.setLayout(null);
38         f.setVisible(true);
39
40         btn.addActionListener(e ->
41             JOptionPane.showMessageDialog(f,
42                 "Registration Successful!\n" +
43                 "Name: " + n.getText() +
44                 "\nRoll No: " + r.getText() +
45                 "\nCourse: " + c.getText()
46             );
47     }
48 }
```



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Output :

Student Registration Form

Name: Atharva Ambilwade

Roll No: 55

Course: Java

Register

Student Registration Form

Message

Registration Successful!

Name: Atharva Ambilwade

Roll No: 55

Course: Java

OK

Register



Q2) Create a simple Employee Registration Form using Java Swing GUI that takes employee details such as Employee ID, Name, Department, and Salary. Display the entered information in a dialog box.

Sol) Code :

```
J Javaswng2.java > ...
1  import javax.swing.*;
2  public class Javaswng2 {
3      Run main | Debug main | Run | Debug
4      public static void main (String[] args) {
5          JFrame frame = new JFrame("Employee Registration Form");
6
7          JLabel idLabel = new JLabel("Employee ID:");
8          JLabel nameLabel = new JLabel("Name:");
9          JLabel deptLabel = new JLabel("Department:");
10         JLabel salaryLabel = new JLabel("Salary (Per annum):");
11
12         JTextField idField = new JTextField();
13         JTextField nameField = new JTextField();
14         JTextField deptField = new JTextField();
15         JTextField salaryField = new JTextField();
16
17         JButton submitButton = new JButton("Submit");
18
19         idLabel.setBounds(40, 40, 100, 30);
20         idField.setBounds(150, 40, 200, 30);
21
22         nameLabel.setBounds(40, 90, 100, 30);
23         nameField.setBounds(150, 90, 200, 30);
24
25         deptLabel.setBounds(40, 140, 100, 30);
26         deptField.setBounds(150, 140, 200, 30);
27
28         salaryLabel.setBounds(40, 190, 100, 30);
29         salaryField.setBounds(150, 190, 200, 30);
30
31         submitButton.setBounds(150, 240, 100, 30);
32
33         frame.add(idLabel);
34         frame.add(idField);
35         frame.add(nameLabel);
36         frame.add(nameField);
37         frame.add(deptLabel);
38         frame.add(deptField);
39         frame.add(salaryLabel);
40         frame.add(salaryField);
41         frame.add(submitButton);
42
43         frame.setSize(420, 350);
44         frame.setLayout(null);
45         frame.setVisible(true);
46
47         submitButton.addActionListener(e -> {
48             JOptionPane.showMessageDialog(
49                 frame,
50                 "Employee Details\n"
51                 + "Employee ID: " + idField.getText()
52                 + "\nName: " + nameField.getText()
53                 + "\nDepartment: " + deptField.getText()
54                 + "\nSalary: " + salaryField.getText()
55             );
56         });
57     }
58 }
```



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Output :

Employee Registration Form

Employee ID: 025

Name: Agam Oze

Department: Finance

Salary (Per ann... 100000

Submit

Employee Registration Form

Message

Employee Details
Employee ID: 025
Name: Agam Oze
Department: Finance
Salary: 100000

OK

Submit